Mahmoud Alyosify

Data Scientist | Predictive Modelling, Experimentation & Decision Analytics
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PROFESSIONAL SUMMARY

Data Scientist who builds models around the **decision they have to support**, not around the leaderboard. Built a satellite collision-risk model deliberately restricted to data available **48+ hours** before closest approach — matching the real operational window — and a reproducible **repository-mining study** answering a measurable question about engineering effort. Strong statistical foundation: six years teaching **probability, statistics, discrete mathematics and graph data mining** to **10,000+ learners**, and research-grade experimentation including **paired non-inferiority testing** and controlled ablation sweeps. As an **AI Engineer Intern at Injaz Tech**, designs the evaluation behind an LLM assistant — retrieval benchmarks, category-split scoring and a CI regression gate. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**. **Microsoft Certified: Data Analyst Associate**.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Deep Learning, Large Language Models, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Quantitative foundation in statistics, probability, discrete mathematics, data analysis and bioinformatics.
- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Statistics & Experimentation	Probability & Statistical Inference, Hypothesis Testing, Paired Non-Inferiority Testing, Controlled Experiments & Ablation Design, Class-Imbalance Handling, Cross-Validation, Error Analysis, LLM & RAG Evaluation (LLM-as-Judge, Retrieval Metrics, MRR), Reproducible Analysis
Modelling	Scikit-learn, XGBoost, LightGBM, Logistic & Linear Models, Feature Engineering, Hyperparameter Optimisation (Optuna), Model Explainability (SHAP), Deep Learning (PyTorch, TensorFlow)
Data Engineering	SQL, Pandas, NumPy, PySpark, ETL Pipelines, Data Cleaning & Validation, Repository & API Data Mining, MySQL, MongoDB, SQL Server
Analytics & Reporting	Power BI (*Microsoft Certified: Data Analyst Associate*), Matplotlib, Streamlit Dashboards, Crystal Reports, Executive-Ready Insight Communication
Deployment	AWS, Docker, FastAPI, ONNX, Git/GitHub, Linux, CI/CD
Programming	Python, SQL, R, C++, C#, MATLAB, Bash

SELECTED DATA SCIENCE PROJECTS

SCRAP — Satellite Collision Risk Assessment & Prediction — *2026*
Python · Scikit-learn · Feature Engineering · Imbalanced Classification

- Built a supervised risk model constrained to conjunction data available **48+ hours** before closest approach — deliberately trading offline accuracy for a prediction that arrives in time to act on.
- Engineered features from raw conjunction records and handled a severely imbalanced target, selecting evaluation metrics appropriate to rare, high-consequence events rather than headline accuracy.

Analyzing Review Effort: Human vs. Agentic Pull Requests — *2026*
Data Mining · Statistical Analysis · Reproducible Pipelines

- Built an end-to-end **GitHub-mining pipeline** — collection, cleaning, cohort definition and analysis — to quantify whether AI-generated pull requests shift reviewer effort compared with human-authored ones.
- Defined comparable cohorts and explicit inclusion criteria so the comparison measured the intended effect rather than sampling artefacts.

SimCLR-Vision-SSL — Controlled 42-Experiment Study — *2026*
PyTorch · Experimental Design · Ablation Analysis

- Designed and executed a **42-experiment sweep** isolating which augmentations drive representation quality, reaching **84.30% top-1 accuracy** — a controlled study rather than a single tuned run.

MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference *2026*
MSc Thesis, Queen's University · Statistical Testing

- Fewer generated tokens are a real saving only when the answer is still complete. **MinimalLM gives both, without training. 47.6%** fewer generated tokens (95% CI 45.7–49.5%) on 330 questions and three open LLMs, with **no answer cut off** (0 of 990); LLM-judge score 8.04/10 against 8.42, and 6.46 for the originals cut to the same length.
- Six-system ablation on 5,940 answers with paired **McNemar** tests and **bootstrap** intervals. The judge was checked against a blind internal reading of 96 answers: Cohen's kappa 0.97 on cut-off answers, only 0.26 on missing parts, so those shares are reported as upper estimates.

Additional Projects

- **ATM Transaction Analysis** — exploratory data analysis and statistical reporting on transaction data in **R**.
- **Diabetes Onset Prediction** — supervised classification pipeline on clinical risk-factor data.
- **Horus-OSINT** — curated **159,826** GTD/GDELT records into a training corpus and deployed a fine-tuned model on AWS for natural-language querying of global event data.
- **Also:** RL-Cloud-Autoscaler (PPO/DQN cost optimisation) · AudioShield · HORUS Sentinel · Vitalism Solution.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Designed the evaluation for an LLM assistant: retrieval benchmarks (top-1, MRR; **75% to 94%**) and category-split scoring with a CI regression gate.
- Used a deterministic key-fact metric to show a +0.06 LLM-judge gain was wording sensitivity, not better answers.

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven analytical tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend data systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**.
- Co-developed an internal **data-management and reporting** application in **C#, .NET, Entity Framework, SQL Server** and **Crystal Reports**, delivering operational reporting across multiple departments.

Online Instructor — Statistics, Probability & Data Science *2020 – Present*
Udemy & YouTube Remote

- Authored and taught Arabic-language courses in **probability, statistics, discrete mathematics, algorithms and graph data mining** for **10,000+ learners** (**700K+ views**).

CERTIFICATIONS & AWARDS

Microsoft Certified: Data Analyst Associate (Power BI)
Machine Learning Specialization — DeepLearning.AI
HCIP – AI (Huawei)
1st Place — Smart Cities Hackathon 2022
AWS Academy — Data Engineering, ML Foundations, ML for NLP
Data Analysis in R — 57357 Children's Cancer Hospital
ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.